

Problem 1 : Basic Exam, Spring 2017

Show that there is a constant C so that

$$\left| \frac{f(0) + f(1)}{2} - \int_0^1 f(x) \, dx \right| \leq C \int_0^1 |f''(x)| \, dx,$$

for every C^2 function $f : \mathbb{R} \rightarrow \mathbb{R}$.

Solution.

$$\begin{aligned} \int_0^1 f(x)g''(x) \, dx &= g'(x)f(x)|_0^1 - \int_0^1 g'(x)f'(x) \, dx \\ &= (g'(1)f(1) - g'(0)f(0)) - \left[[g(x)f'(x)]_0^1 - \int_0^1 g(x)f''(x) \, dx \right] \\ &= g'(1)f(1) - g'(0)f(0) - g(1)f'(1) + g(0)f'(0) + \int_0^1 g(x)f''(x) \, dx \\ \int_0^1 g(x)f''(x) \, dx &= -g'(x)f(1) + g'(0)f(0) + g(1)f'(1) - g(0)f'(0) - \int_0^1 f(x)g''(x) \, dx \end{aligned}$$

We want

$$\begin{aligned} g'(1) &= -\frac{1}{2} & g'(0) &= \frac{1}{2} \\ g(1) &= 0 & g(0) &= 0 \\ g''(x) &= -1 \end{aligned}$$

Thus

$$\begin{aligned} g'(x) &= -x + A = -x + \frac{1}{2} \\ g(x) &= -\frac{1}{2}x^2 + \frac{1}{2}x = -\frac{1}{2}x(x-1). \end{aligned}$$

Note $|g(x)| \leq 1$. Thus

$$\begin{aligned} \left| \frac{f(0) + f(1)}{2} - \int_0^1 f(x) \, dx \right| &= \left| \int_0^1 \left(-\frac{1}{2}x(x-1) \right) f''(x) \, dx \right| \\ &\leq \int_0^1 |f''(x)| \, dx \end{aligned}$$

Problem 2 : Basic Exam, Fall 2020

Let f be real and continuous on $[0, 1]$. Show that

$$\lim_{n \rightarrow \infty} (n+1) \int_0^1 x^n f(x) \, dx = f(1).$$

Solution. f continuous on $[0, 1]$. Thus $\exists M, m$ such that

$$M = \sup_{x \in [0, 1]} f(x)$$

$$m = \inf_{x \in [0, 1]} f(x)$$

By continuity of f at 1, $\forall \varepsilon > 0$ there exists a $0 < \delta$ where $M(1 - \delta)^{n+1} < \varepsilon$ and $\forall y \in (1 - \delta, 1]$ we have that $|f(1) - f(y)| < \varepsilon$. Thus

$$\begin{aligned} \left| \int_0^1 (n+1)x^n f(x) \, dx - \int_0^1 (n+1)x^n f(1) \, dx \right| &= \left| \int_0^1 (n+1)x^n (f(x) - f(1)) \, dx \right| \\ &= \left| \int_0^{1-\delta} (n+1)x^n (f(x) - f(1)) \, dx + \int_{1-\delta}^1 (n+1)x^n f(x) \, dx \right| \\ &\leq \left| \int_0^{1-\delta} (n+1)x^n (f(x) - f(1)) \, dx \right| + \varepsilon [x^{n+1}]_{1-\delta}^\delta \\ &\leq M[x^{n+1}]_0^{1-\delta} + \varepsilon [x^{n+1}]_{1-\delta}^1 \\ &< \varepsilon + M(1 - \delta)^{n+1} < 2\varepsilon \end{aligned}$$

Problem 3 : Basic Exam, Spring 2015

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a Lipschitz continuous function, that is

$$L(f) := \sup \left\{ \frac{|f(x) - f(y)|}{|x - y|}; x, y \in \mathbb{R}, x \neq y \right\} < \infty.$$

Assume that we have for every $x \in \mathbb{R}$,

$$\lim_{n \rightarrow \infty} n \left(f \left(x + \frac{1}{n} \right) - f(x) \right) = \lim_{n \rightarrow \infty} n \left(f \left(x - \frac{1}{n} \right) - f(x) \right) = 0.$$

Show that f is differentiable on $\mathbb{R}$.

Solution. It suffices to prove the statement for sequences of real numbers which converge at 0 and are positive and negative. Suppose $a_n \rightarrow 0$ is a sequence of positive numbers. Let $m_n = \left\lceil \frac{1}{a_n} \right\rceil$ is a subsequence of integers. Thus $1 \leq a_n m_n \leq 1 + a_n \rightarrow 1$. Thus there exists a $N_0 \in \mathbb{N}$ such that $|1 - a_n m_n| < \frac{\varepsilon}{2L(f)}$. By assumption, there exists a $N_1 \in \mathbb{N}$ such that for all $n \geq N$ we have that

$$\left| n \left(f \left(x + \frac{1}{n} \right) - f(x) \right) \right| < \frac{\varepsilon}{2}.$$

There exists a $M_0 \in \mathbb{N}$ such that $a_n < \left(\frac{1}{M+1} \right)$ for all $n \geq M_0$. Thus for all $n \geq \max(N_0, M_1)$:

$$\begin{aligned} \left| \frac{f(x + a_n) - f(x)}{a_n} \right| &\leq \left| \frac{f \left(x + \frac{1}{m_n} \right) - f(x)}{a_n} \right| + \left| \frac{f \left(x + \frac{1}{m_n} \right) - f(x - a_n)}{a_n} \right| \\ &\leq \frac{\varepsilon}{2} \left| \frac{1}{m_n a_n} \right| + L(f) \left| \frac{\frac{1}{m_n} - a_n}{a_n} \right| \\ &\leq \frac{\varepsilon}{2} + L(f) |1 - m_n a_n| \\ &< \varepsilon \end{aligned}$$